

Mixed Reality Racing: Combining Real and Virtual Motorsport Racing

Hsueh Han Wu

Rakuten Mobile, Inc., Setagaya, Tokyo, Japan
hsuehhan.wu@rakuten.com

Kelvin Cheng

Rakuten Institute of Technology, Rakuten Group, Inc.,
Setagaya, Tokyo, Japan
kelvin.cheng@rakuten.com

Madoka Inoue

Rakuten Mobile, Inc., Setagaya, Tokyo, Japan
madoka.inoue@rakuten.com

Soh Masuko*

Shibaura Institute of Technology, Koto, Tokyo, Japan
masukon@shibaura-it.ac.jp



Figure 1: (a) (b) An overview of our mixed reality racing prototype. A real race car (blue/white) races against a virtual game car (pink) in real time. (c) Audience at the grandstand, using our AR system to watch the real race car competing with the virtual car. (d) A screenshot of the AR app, where the virtual car and its telemetry information were shown in AR. (e) The racing simulator was located remotely, and audience there could watch the race on the projection screen.

ABSTRACT

In this work, we proposed a proof-of-concept system that combines real and virtual racing, allowing professional racing drivers and e-racers to compete in real time. The real race car is imported into a racing game, and the virtual game car is exported into the real world in AR. This allows e-racers to compete against professional racing drivers and gives the audience an exciting experience from the grandstand with the use of AR. Our system was deployed during an actual racing event, and we gathered initial feedback from users. Both e-racers and audience expressed excitement about our system, indicating a strong potential for this new form of mixed reality racing.

ACM Reference Format:

Hsueh Han Wu, Kelvin Cheng, Madoka Inoue, and Soh Masuko. 2023. Mixed Reality Racing: Combining Real and Virtual Motorsport Racing. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH '23 Posters)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588028.3603645>

1 INTRODUCTION

Motor racing has been a popular sporting competition throughout history, and in recent years interest in e-sports has grown tremendously. It begs the question if it is possible to let professional e-racers compete with professional drivers in their own respective

machines. This motivated us to propose a novel “Mixed Reality Racing” experience that fuses real and virtual racing simultaneously. In recent years, there has been an increase in mixed reality research, but scenarios involving racing are rather limited, mostly related to low-speed driving tasks, passenger-oriented tasks, or non-driving tasks [Ghiurău et al., 2020] [Hock et al., 2017]. In terms of commercial projects, “Honda R vs. R,” reproduced an AR car from a pre-recorded racing game, and had the real driver compete with this “ghost” AR car shown on the windscreen. In our project, we focused on the opposite, which is to let an e-racer compete with a real driver within a game. And more importantly, we propose a real-time racing system, where real and virtual race cars can compete in real time. We achieved this by transmitting the location information of the real race car into a racing game, where it appears as a “digital twin” version of the real car within the game (Figure 1b). On the other hand, we created a dedicated mobile AR app for the on-site audience at the track for them to watch the virtual game car in AR competing with the real race car (Figure 1c and 1d). To achieve this, the virtual game car position was extracted from the game server and combined with a VPS (Visual Positioning System), allowing us to accurately place a “physical twin” version of the game car directly on the real race track, at the same position corresponding to the virtual race track.

2 IMPLEMENTATION

Our system (Figure 2) was implemented at the 2022 Inter Proto Series POWERED BY KeePer Round 2 race held at Fuji Speedway in Shizuoka Prefecture, Japan. On the other hand, the racing simulator was placed remotely at the Rakuten Mobile Ebisu Store in central Tokyo. 5G mmWave was used for communication between the racing simulator and the game server, thereby linking the real and virtual cars, allowing them to race simultaneously. Our system comprises of two sub-systems related to two user experiences:

*Previously affiliated with Rakuten Group, Inc.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Posters, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0152-8/23/08.

<https://doi.org/10.1145/3588028.3603645>

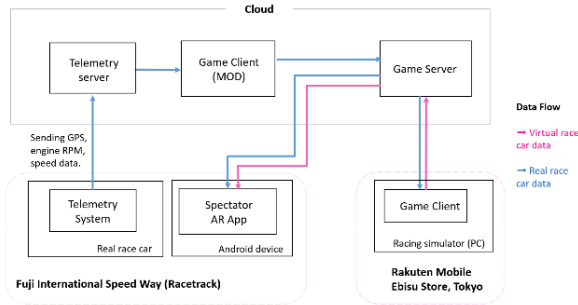


Figure 2: The data communication including both the racing simulator and the spectator AR application.

Driving experience. The first system allowed e-racers to compete against a physical race car in a racing simulator. We used a telemetry system and a telemetry server, both provided by M-TEC Co., Ltd. The telemetry system was installed in the real race car, and transmitted sensor data such as GPS, speed, RPM to the telemetry server on the cloud. AssettoCorsa, a racing game was used in our system. In one of the game clients, we developed an extension or "MOD" that converted the real physical race car data into game car data, which was then synchronized with the game server. Within the game, a replica of the real car with the same colour and appearance was created to represent the real car in the game. In addition, a 3D replica of the real race track was used, resulting in a 1-to-1 correspondence between the real track and the virtual track. As a result, the car can be positioned on the virtual track at the exact same location as on the real track.

Spectating experience in AR. The second system allowed audience to watch the game car, driven remotely by an e-racer, compete with the real race car, from the grandstand. The system was developed with Unity and deployed on Android mobile phones. Game car data coming from the racing simulator, such as position, speed, RPM, were extracted from the game server and streamed to the AR app in real time. We used a VPS provided by Immersal Ltd. to accurately place the AR car on the actual race track corresponding to the game car position in the virtual race track. In addition, both the real race car and virtual game car telemetry data were shown in the app in the form of Head-Up Display (HUD) to provide more immersive and informative experience. Since the audience in the grandstand could only see a portion of the track, the app also included live video streaming, allowing the audience to see the race from the game view, as well as on-board camera from the real race car, while the race cars were out of sight.

3 PRELIMINARY USER STUDY

We collected user feedback through questionnaires from three different groups of users: e-racers, racing simulator audience, and AR audience in the grandstand. The number of the participants in each group was 3, 8, and 10, respectively, and selected results are shown in Figure 3. Our primary focus was on the hedonic qualities of this new racing experience, so we used Q1, Q3, and Q5 to assess each group's excitement. The results were very positive and almost all participants reporting feeling excited about this racing experience, whether they were driving or spectating. We also elicited feedback on the experience of racing between real and virtual cars. Q2 aimed

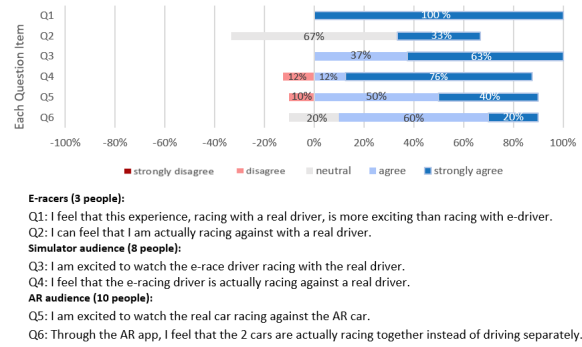


Figure 3: The selected questions from our questionnaires, and the results are shown with diverging stacked bar chart.

to understand if e-racers felt they were competing against a real driver. While two e-racers responded "neutral" to this question, one "strongly agree"-d. This participant explained that he "felt the opponent driving like a real driver because of the particular driving style". This suggests that were it not for an experienced e-racer, it might be difficult for an amateur user to tell the difference between real and virtual cars. We also asked the simulator and AR audience whether they felt that the e-racer and the real driver were racing together. The AR app audience generally agreed, but for the simulator audience, this feeling was amplified, with 6 out of 8 of them strongly agreeing. This could be due to the fact that the simulator audience could see the duel being battled out over the course of the lap, while grandstand users could only see a small part of it.

Overall, our system received positive feedback regarding both the driving and viewing experiences, demonstrating that we achieved our primary goal of creating excitement through this new form of racing experience. As future work, it might be necessary to devise ways to further distinguish virtual cars that are linked with real cars racing on the real race track.

4 CONCLUSION AND FUTURE WORK

In this work, we have successfully implemented a proof-of-concept system that seamlessly integrated real-world and virtual racing in real time. E-racers were able to compete against a car running on a real race track, while the audience in the grandstand were able to watch the mixed reality race through an AR app. Although the sample size for the driving experience was limited, initial feedback indicated positive results for both the driving and viewing experiences. However, to fully complete the experience, it would be necessary to explore potential solutions for displaying the position of the e-racer's car in the real car driver's view to create a more exciting experience.

REFERENCES

- Florin-Timotei Ghiurău, Mehmet Aydın Baytaş, and Casper Wickman. 2020. ARCAR: On-Road Driving in Mixed Reality by Volvo Cars. In Adjunct Publication of the 33rd Annual ACM Symposium on User Interface Software and Technology. Association for Computing Machinery, 62–64. DOI: <https://doi.org/10.1145/3379350.3416186>
- Philipp Hock, Sebastian Benedikter, Jan Gugenheimer, and Enrico Rukzio. 2017. CarVR: Enabling In-Car Virtual Reality Entertainment. ACM Press, 4034–4044. DOI: <http://dx.doi.org/10.1145/3025453.3025665>
- UNIT9. 2020. Honda R vs R - Mixed Reality Racing. Retrieved April 10, 2023 from <https://www.behance.net/gallery/92189065/HONDA-R-vs-R-Mixed-Reality-RacingTables>